

Assignment 5: Schema creation and constraints

Statement: Create the schema and constraints on the relations.

Modify the trains schema which we saw earlier to create constraints to check the following:

- The value of timein is always less than or equal to timeout

→

```
ALTER TABLE trainhalts
ADD CONSTRAINT chk_time
CHECK (timein <= timeout);
```

```
MYSQL root@localhost (railway) > ALTER TABLE trainhalts
-> ADD CONSTRAINT chk_time
-> CHECK (timein <= timeout);
16:14 [10/1667]
You're about to run a destructive command.
Do you want to proceed? (y/n): y
Your call!
Query OK, 19 rows affected
Time: 0.068s
Mysql root@localhost (railway) > SHOW CREATE TABLE train;
+-----+
| Table | Create Table |
+-----+
| train | CREATE TABLE `train` (
|       |   `id` varchar(5) NOT NULL,
|       |   `name` varchar(20) DEFAULT NULL,
|       |   PRIMARY KEY (`id`)
|       | ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
1 row in set
Time: 0.004s
Mysql root@localhost (railway) > SHOW CREATE TABLE trainhalts;
+-----+
| Table      | Create Table |
+-----+
| trainhalts | CREATE TABLE `trainhalts` (
|           |   `id` varchar(5) NOT NULL,
|           |   `seqno` int NOT NULL,
|           |   `stcode` varchar(10) DEFAULT NULL,
|           |   `timein` varchar(5) DEFAULT NULL,
|           |   `timeout` varchar(5) DEFAULT NULL,
|           |   PRIMARY KEY (`id`,`seqno`),
|           |   CONSTRAINT `chk_time` CHECK ((`timein` <= `timeout`))
|           | ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
```

- When a train is removed from service, all its halts should be deleted.

```
ALTER TABLE trainhalts ADD CONSTRAINT fk_trainhalts_train FOREIGN KEY
(id) REFERENCES train(id) ON DELETE CASCADE;
```

```
INSERT INTO train VALUES ('T20', 'Maharashtra Express');
```

```
INSERT INTO trainhalts (id, seqno, stcode, timein, timeout) VALUES
('T20', 999, 'PUNE', '12:00', '13:00');
```

```
DELETE FROM train WHERE id = 'T20'
```

```

MYSQL root@localhost (railway) > INSERT INTO trainhalts
-> (id, seqno, stcode, timein, timeout)
-> VALUES
-> ('T20', 999, 'PUNE', '12:00', '13:00');

Query OK, 1 row affected
Time: 0.008s
MYSQL root@localhost (railway) > DELETE FROM train
-> WHERE id = 'T20';

You're about to run a destructive command.
Do you want to proceed? (y/n): y
Your call!
Query OK, 1 row affected
Time: 0.008s
MYSQL root@localhost (railway) > SELECT *
-> FROM trainhalts
-> WHERE id = 'T20';

+-----+-----+-----+-----+-----+
| id | seqno | stcode | timein | timeout |
+-----+-----+-----+-----+-----+

0 rows in set
Time: 0.003s

```

- Insert inconsistent data and verify the constraints.

```

MYSQL root@localhost (railway) > INSERT INTO trainhalts
-> (id, seqno, stcode, timein, timeout)
-> VALUES
-> ('T1', 999, 'PUNE', '12:00', '10:00');
(3819, "Check constraint 'chk_time' is violated.")
MYSQL root@localhost (railway) > INSERT INTO trainhalts
-> (id, seqno, stcode, timein, timeout)
-> VALUES
-> ('T1', 999, 'PUNE', '12:00', '13:00');
(1452, 'Cannot add or update a child row: a foreign key constraint fails (`railway`.`trainhalts`, CONSTRAINT `fk_trainhalts_train` FOREIGN KEY (`id`) REFERENCES `train` (`id`) ON DELETE CASCADE)')
MYSQL root@localhost (railway) > insert into train (id, name) values ('T1', 'maharashtra express');
Query OK, 1 row affected
Time: 0.007s
MYSQL root@localhost (railway) > INSERT INTO trainhalts
-> (id, seqno, stcode, timein, timeout)
-> VALUES
-> ('T1', 999, 'PUNE', '12:00', '13:00');

Query OK, 1 row affected
Time: 0.006s
MYSQL root@localhost (railway) > select * From trainhalts where id = 'T1';

+-----+-----+-----+-----+-----+
| id | seqno | stcode | timein | timeout |
+-----+-----+-----+-----+-----+
| T1 | 999 | PUNE | 12:00 | 13:00 |
+-----+-----+-----+-----+-----+

1 row in set
Time: 0.004s
MYSQL root@localhost (railway) > DELETE FROM train
-> WHERE id = 'T1';

You're about to run a destructive command.
Do you want to proceed? (y/n): y
Your call!
Query OK, 1 row affected

```

```

Time: 0.000s
MYSQL root@localhost (railway) > select * From trainhalts where id = 'T1';
+-----+-----+-----+-----+-----+
| id | seqno | stcode | timein | timeout |
+-----+-----+-----+-----+-----+
| T1 | 999 | PUNE | 12:00 | 13:00 |
+-----+-----+-----+-----+-----+

1 row in set
Time: 0.004s
MYSQL root@localhost (railway) > DELETE FROM train
                                -> WHERE id = 'T1';
You're about to run a destructive command.
Do you want to proceed? (y/n): y
Your call!
Query OK, 1 row affected
Time: 0.006s
MYSQL root@localhost (railway) > SELECT *
                                -> FROM trainhalts
                                -> WHERE id = 'T1';
+-----+-----+-----+-----+-----+
| id | seqno | stcode | timein | timeout |
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+

0 rows in set
Time: 0.003s
MYSQL root@localhost (railway) >

```

Write SQL Create table statements to create the following schema. Include all appropriate primary and foreign key declarations. Choose appropriate types for each attribute.

- remotecentre(centreId, college, town, state)
- person(ID, name, email)
- programme(progId, title, fromdate, todate)
- coordinator(ID, progId, centreId)
- participant(ID, progId, centreId)

```

→
CREATE TABLE remotecentre (
    centreId INT PRIMARY KEY,
    college VARCHAR(100) NOT NULL,
    town VARCHAR(50),
    state VARCHAR(50)
);

```

```

CREATE TABLE person (
    ID INT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE
);

```

```

CREATE TABLE programme (
    progId INT PRIMARY KEY,
    title VARCHAR(100) NOT NULL,
    fromdate DATE,
    todate DATE,
    CHECK (fromdate <= todate)
);

```

```
);
```

```
CREATE TABLE coordinator (  
    ID INT,  
    progId INT,  
    centreId INT,  
  
    PRIMARY KEY (ID, progId, centreId),  
  
    FOREIGN KEY (ID)  
        REFERENCES person(ID),  
  
    FOREIGN KEY (progId)  
        REFERENCES programme(progId),  
  
    FOREIGN KEY (centreId)  
        REFERENCES remotecentre(centreId)  
);
```

```
CREATE TABLE participant (  
    ID INT,  
    progId INT,  
    centreId INT,  
  
    PRIMARY KEY (ID, progId, centreId),  
  
    FOREIGN KEY (ID)  
        REFERENCES person(ID),  
  
    FOREIGN KEY (progId)  
        REFERENCES programme(progId),  
  
    FOREIGN KEY (centreId)  
        REFERENCES remotecentre(centreId)  
);
```

```
-> centreId INT,  
->  
-> PRIMARY KEY (ID, progId, centreId),  
->  
-> FOREIGN KEY (ID)  
-> REFERENCES person(ID),  
->  
-> FOREIGN KEY (progId)  
-> REFERENCES programme(progId),  
->  
-> FOREIGN KEY (centreId)  
-> REFERENCES remotecentre(centreId)  
-> );  
->
```

Query OK, 0 rows affected
Time: 0.029s

Query OK, 0 rows affected
Time: 0.026s

Query OK, 0 rows affected
Time: 0.022s

Query OK, 0 rows affected
Time: 0.035s

Query OK, 0 rows affected
Time: 0.037s

MYSQL root@localhost (college_program) >